



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2011

BIOLOGY
Higher 2



Friday
12 August 2011

1 hour 15 minutes

Additional materials:
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Write your name, CG and index number in the spaces at the top of this page.

Enter your name, subject title, test name, class. For your index number, enter your full NRIC number. Shade the corresponding lozenges on the OMS according to the instructions given by the invigilators.

There are forty questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C, D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

AT THE END OF THE EXAMINATION, HAND IN BOTH OMS AND QUESTION PAPERS.

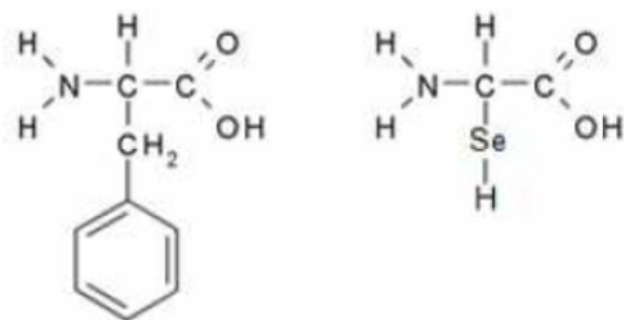
INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done on the question paper.

This question paper consists of 20 printed pages.

1. The diagram shows two molecules of a biological substance.



Which type of bond is formed when these molecules react to form a larger molecule?

- A Phosphodiester
- B Glycosidic
- C Peptide
- D Hydrogen

2. Which one of the following statements is **INCORRECT**?

- A There are 3 polypeptide chains in the collagen helix, and only one in the α - helix
- B Proline is a major component of collagen but not of the α -helix.
- C Every third residue in the α -helix must be glycine, but there is no sequence regularity in collagen
- D Some hydrogen bonding exists between different chains in collagen, but hydrogen bonding occurs within the same chain in the α -helix.

3. Which of the following is/ are true of triglycerides and phospholipids?

- i. Both contain fatty acid tails
- ii. Only phospholipids are amphipathic
- iii. Phospholipids are formed from fatty acids and phosphoric acid while triglycerides are formed from fatty acids and glycerol.
- iv. Both fatty acids and glycerol are soluble in organic solvents
- v. Phospholipids have a higher carbon content than triglycerides for storage purposes

- A i, ii and iii
- B i, ii and iv
- C ii, iii and iv
- D ii, iii and v

4. A pear was cut into six equal segments, two of which were blanched by immersing them in boiling water for one minute. Each segment of the pear was then cut into pieces and these were treated in several ways as indicated in the table below. The time taken for each piece of pear to develop a standard degree of browning was then recorded. You may assume that all pieces had the same initial colour

Liquid in which pear pieces were immersed	Initial treatment of pear segment	Time for appearance of browning at the temperature indicated (min)		
		20°C	40°C	100°C
Water	Unblanched	25	9	X
	blanched	X	X	X
1% glucose solution	Unblanched	X	X	5
	blanched	X	X	4
0.1M nitric acid	Unblanched	X	X	X
2% dithionite solution	Unblanched	X	135	X

X indicates that the piece of pear did not turn brown within 180 minutes.

The enzyme involved in the browning of the pear pieces is

I thermostable.

II destroyed by blanching.

III inhibited more by 2% dithionite solution than by 0.1M nitric acid.

IV is not the only agent responsible for the browning of the pear in the experiment.

A **I** and **III** only

B **II** and **III** only

C **II** and **IV** only

D **II**, **III** and **IV** only

5. Enzyme molecules with high optimum temperature will have

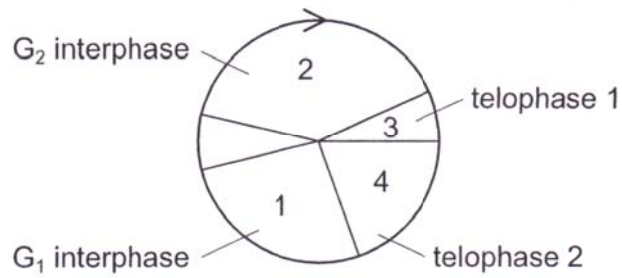
A greater number of peptide bonds than other enzyme molecules.

B greater number of cysteine residues in their polypeptide chains.

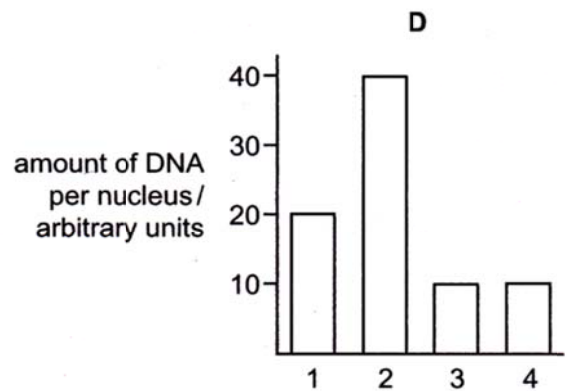
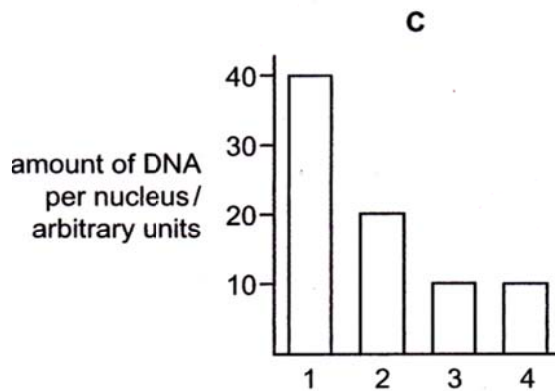
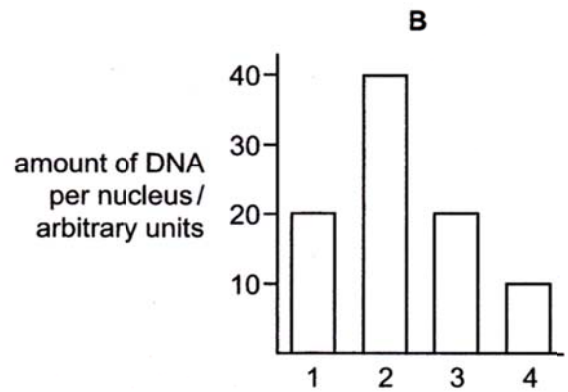
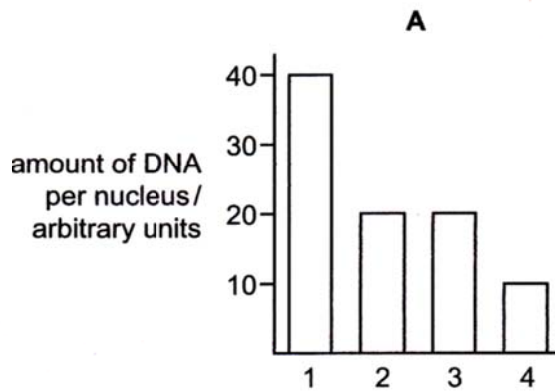
C greater number of hydrophobic amino acid residues.

D greater number of associated non-protein groups.

6. The amount of DNA per nucleus in a cell during a meiotic cell cycle, as shown below, was measured.

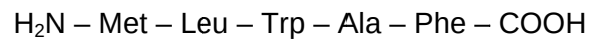


Which bar graph correctly represents the variation in the DNA content?



7. During gametogenesis, non-disjunction occurs in one daughter cell during meiosis II. What would be the result at the completion of meiosis?
- A** All gametes would be haploid.
 - B** Two gametes would be haploid and two will be diploid.
 - C** Two gametes would have (n+1) chromosomes and two would have (n-1) chromosomes.
 - D** One gamete would have (n+1) chromosomes, one would have (n-1) chromosomes and the other two will have n chromosomes.
8. In a hypothetical species of frogs, there are 6 tetrads at metaphase of meiosis I. How many chromosomes would be present in each cell at the end of meiosis II?
- A** 3
 - B** 6
 - C** 12
 - D** 24
9. Identify the correct statements regarding the replication of DNA
- 1 Only 1 strand of DNA is replicated
 - 2 It occurs in S phase of interphase in both mitosis and meiosis
 - 3 It occurs after the completion of telophase in meiosis
 - 4 DNA polymerase III is required (for bacteria)
- A** 1 and 3 only
 - B** 2 and 4 only
 - C** 2 and 3 only
 - D** 2 only
10. The amino acid pyrrolysine is an unusual amino acid present in the proteins of some methane producing organisms. Pyrrolysine is encoded in mRNA by UAG, which is the 'amber' stop codon in most organisms.
- Which of the following is/are required in order for pyrrolysine to be incorporated into polypeptides?
- i. Specific tRNA molecule
 - ii. Specific peptidyl transferase
 - iii. Specific aminoacyl-tRNA synthetase
 - iv. Specific small ribosomal subunits
- A** i and iii
 - B** i and iv
 - C** ii and iii
 - D** ii and iv

11. A pentapeptide has the following amino acid sequence:



where

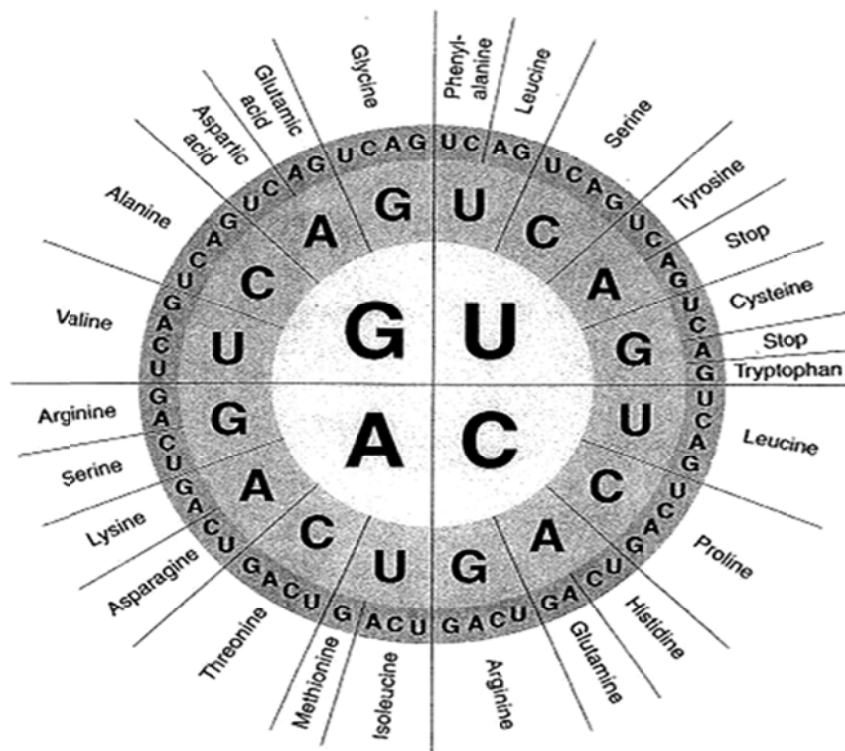
Met : methionine

Ala : alanine

Leu : leucine

Phe : phenylalanine

Trp : tryptophan



Using the genetic code shown in the diagram above, deduce the corresponding template strand sequence during transcription:

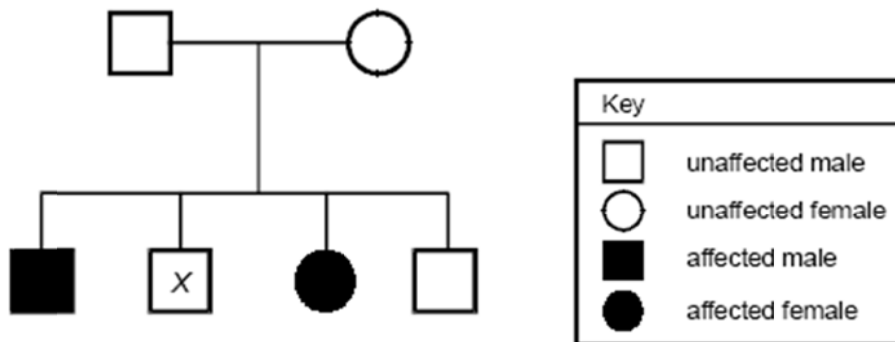
- A 5' – ATG TTG TGG GCC TTT TAG – 3'
- B 5' – AUG CUG UGG GCA UUC – 3'
- C 5' – TTA GAA GGC CCA CAA CAT – 3'
- D 5' – GAA GGC CCA CAA CAT – 3'

12. Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein product of that gene?

	Promoter	Terminator	Start codon	Stop codon	Middle of an intron
A	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
B	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced

B

13. The family pedigree above shows the inheritance of a trait that is determined by two alleles. What is the probability that the individual labeled X is homozygous?



- A 25%
 B 33%
 C 50%
 D 67%

14. Two true breeding *Drosophila* are crossed: a normal-winged, red-eyed female and a miniature-winged, vermilion-eyed male. The F1 all have normal wings and red eyes. F1 offspring are crossed with miniature-winged, vermilion-eyed flies. The following offspring were obtained:

233 normal wing, red eye
 247 miniature wing, vermilion eye
 7 normal wing, vermilion eye
 13 miniature wing, red eye

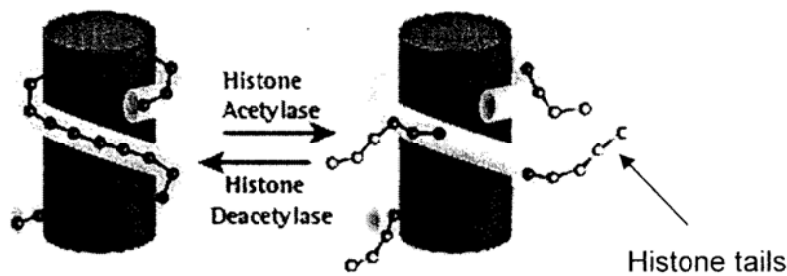
From these data, one could conclude that the alleles for miniature wings and vermilion eye are

- A** both X-linked and dominant.
B located on autosomes and dominant
C recessive, and these genes are located 4 map units apart.
D recessive, and the deviation from the expected 9:3:3:1 ratio is due to epistasis.

15. Which combination of factors will affect translational control of gene expression?

	presence of 5' cap on mRNA	presence of carbohydrate group attached to polypeptide	presence of poly(A) tail at 3' end of mRNA	presence of repressor bound to 5' UTR of mRNA	presence of ubiquitin attached to polypeptide
A	no	yes	no	no	yes
B	no	yes	no	yes	yes
C	yes	no	yes	yes	no
D	yes	yes	yes	yes	yes

16. As shown in the diagram below, acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modifications of histones play a direct role in regulation of gene transcription. Suggest how this works.



- A Helicase action is enhanced with acetylation of the histones.
 B Acetylation of histone tails neutralizes their negative charges and encourage binding of DNA polymerase I.
 C When nucleosomes are highly acetylated, the chromatin becomes less compact and DNA is more accessible for transcription.
 D RNA polymerase work better by binding to acetyl groups.
17. In eukaryotes, many genes may have to interact with each other, requiring more regulatory proteins that can fit around a single promoter. This physical limitation is overcome by
- A The use of very long promoters.
 B Alternating promoters and operators.
 C Distant sites in a chromosome controlling transcription of a gene
 D Having factors on one chromosome control genes on another gene
18. Which combination is correct?

	ATP used	Dehydrogenation	Reduced NADP used
A	Calvin cycle	glycolysis	Krebs cycle
B	glycolysis	Calvin cycle	Krebs cycle
C	glycolysis	Krebs cycle	Calvin cycle
D	Krebs cycle	glycolysis	Calvin cycle

19. The blue dye DCPIP can be converted to reduced DCPIP by gaining electrons. This is summarised below.



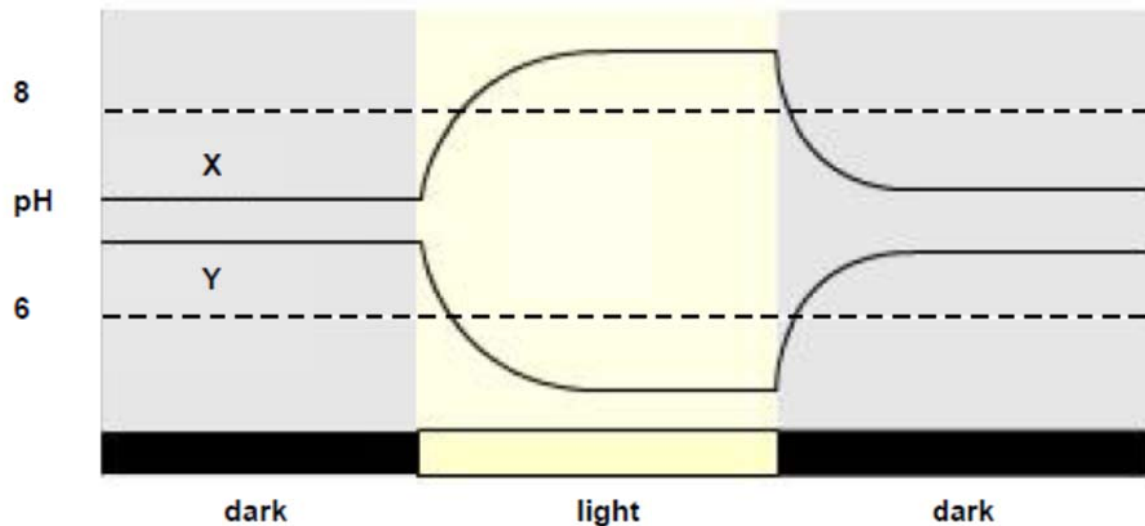
Tube	Contents	Treatment	Colour of solution	
			At start	After 20 min
1	1 cm ³ chloroplast suspension and 5 cm ³ DCPIP	Illuminated strongly	Blue green	Green
2	1 cm ³ buffer solution and 5 cm ³ DCPIP	Illuminated strongly	Blue	Blue
3	1 cm ³ chloroplast suspension and 5 cm ³ DCPIP	Left in the dark	Blue green	Blue green

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following way.

Which one of the following statements is a possible conclusion for the observations above?

- A** Light reaction which occurs in the chloroplasts yield free electrons which reduced DCPIP.
- B** Either strong illumination or the buffer used in extraction of chloroplasts could oxidise DCPIP
- C** Photolysis of water produces oxygen which oxidises DCPIP.
- D** Respiration consumes oxygen from the suspension, hence DCPIP is partially reduced.

20. The diagram below consists of two graphs showing the changes in pH of two structures X and Y, of a chloroplast in the absence and presence of light. Which of the following correctly identifies structures X and Y?



	X	Y	Reason
A	Thylakoid space	Stroma	Protons are pumped from the thylakoid space into the stroma, hence decreasing the latter's pH
B	Stroma	Thylakoid space	Protons are pumped from the stroma into the thylakoid space hence decreasing the latter's pH
C	Thylakoid membrane	Stroma	Protons are pumped from the thylakoid membrane into the stroma, hence increasing the latter's pH
D	Stroma	Thylakoid membrane	Protons are pumped from the stroma into the thylakoid membrane, hence increasing the latter's pH

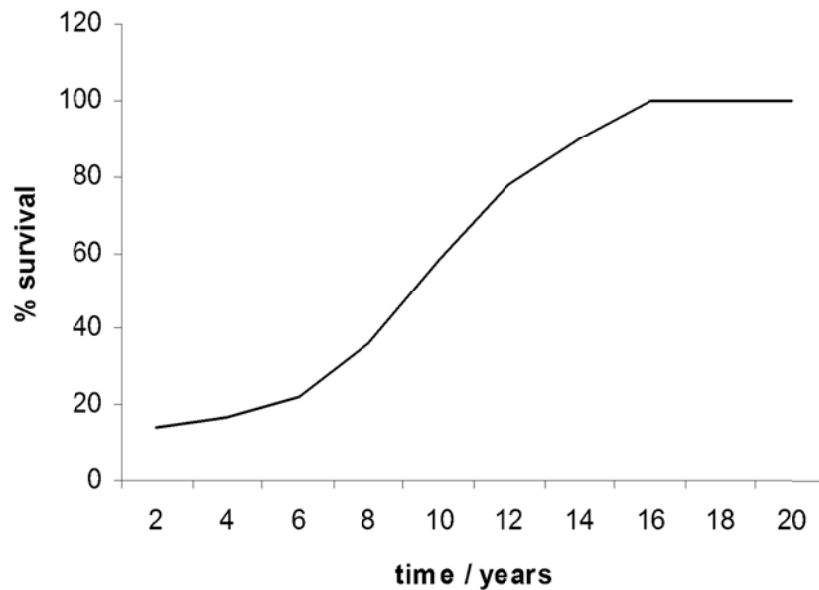
21. The followings are possible events that may lead to speciation.

1. Gene mutations
2. Increased gene flow
3. Natural selection
4. Geographical isolation (e.g. mountain or river)
5. Habitat differentiation within the same geographical location

The correct order of events that may lead to sympatric speciation is

- A 4, 1, 3
- B 5, 1, 3
- C 4, 2, 3
- D 5, 2, 3

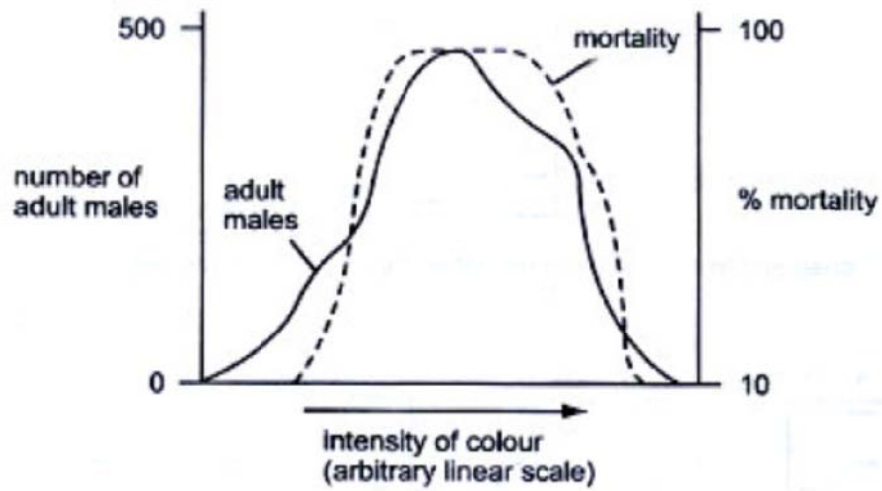
22. The graph shows the effect of pesticide treatment on beetles in Tanzania over a number of years. A standard amount of pesticide was used each year in summer.



How is the effect of pesticide best explained?

- A** A few resistant beetles reproduced more successfully and the resistance allele increased in frequency.
- B** At every generation, an increasing proportion of beetles mutated to become resistant.
- C** Repeated exposure to the pesticide caused the beetles to become more resistant.
- D** The allele for resistance mutated from the recessive form to the dominant form.

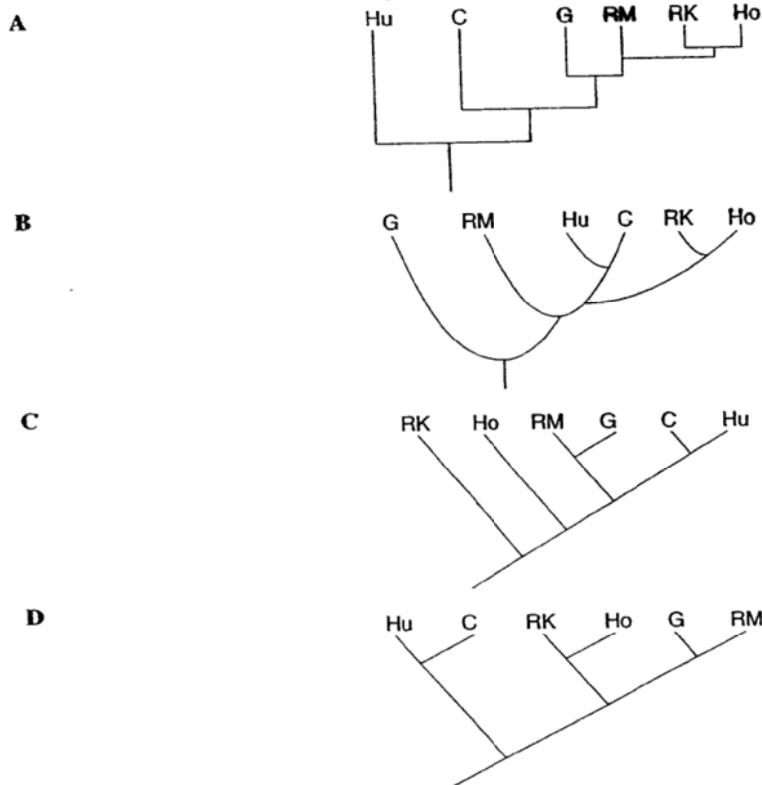
23. The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity. Which conclusion can be made from this graph?



- A Colour variation is environmentally induced.
- B Colour variation is genetically determined.
- C Extreme forms are favoured by natural selection.
- D The species shows discontinuous variation with respect to colour.

24. The table below shows the amino acid sequence of part of the beta-haemoglobin molecule found in 6 different vertebrates. The beta-globin chain has a total of 146 amino acids. Only the amino acids found in positions 101 to 116 are represented.

amino acid position	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116
human (Hu)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
red kangaroo (RK)	glu	asn	phe	lys	leu	leu	gly	asn	ile	ile	val	ile	cys	leu	ala	glu
rhesus monkey (RM)	glu	asn	phe	lys	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
gorilla (G)	glu	asn	phe	lys	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
chimpanzee (C)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	val	cys	val	leu	ala	his
horse (Ho)	glu	asn	phe	arg	leu	leu	gly	asn	val	leu	ala	leu	val	val	ala	arg



25. Epidermal growth factor receptor (EGFR) is encoded by a proto-oncogene. Which genetic change would convert the proto-oncogene into an oncogene?
- A amplification of the promoter that controls the proto-oncogene
 - B amplification of the proto-oncogene
 - C point mutation in an intron within the proto-oncogene
 - D translocation of the proto-oncogene to a region downstream of an inactive promoter
26. Which of the following is **not** a factor that can increase the chance of cancerous growth?
- A infection by influenza virus
 - B inheritance of p53 gene mutation
 - C smoking after every meal
 - D sun tanning every weekend
27. In the lac operon model for lactose metabolism in prokaryotes, what is active when the operon is turned off?
- A Regulatory gene
 - B RNA polymerase
 - C Promoter region
 - D Structural genes
28. Which of the following occurs in the reproductive cycle of the human immunodeficiency virus (HIV) but not in that of the influenza virus?
- A Host cell DNA is hydrolysed.
 - B Host cell ribosomes are used to synthesize viral proteins.
 - C Viral RNA acts as template for viral DNA synthesis.
 - D Viruses enter the host cell by endocytosis.
29. In Generalised Transduction, defective virus are formed as a result of;
- A viral enzymes cutting the host DNA such that the host DNA is assembled into the new virus.
 - B production of host enzymes by virus which nicks its own DNA such that it can be assembled into the new virus.
 - C no shut down of host DNA production such that host DNA or the virus DNA can either be assembled into the new virus.
 - D integrate of virus DNA into host DNA and during excision the viral genome carries along with it the host DNA to be assembled into the new virus.

30. Which of the following viral structures are not completely coded by the virus' own nucleic acids?
- A Capsid
 - B Glycoprotein of HIV
 - C Viral envelope
 - D Tail sheath of T4 bacteriophage
31. Which of the following correctly describes the different components of a homeostatic mechanism for glucose regulation?

	stimulus	receptor	control centre	signal	effector
A	High glucose	islets of Langerhans	liver	glucagon	liver; muscles
B	High glucose	islets of Langerhans	islets of Langerhans	insulin	liver, muscles
C	lack of glucose	islets of Langerhans	islets of Langerhans	insulin	glucose
D	lack of glucose	liver	islets of Langerhans	glucagon	glucose

32. Which of the following is/are true about G-protein signaling?
- I. Testosterone hormone binds to its cell surface receptor to cause GTP to displace GDP, thus activating G-protein.
 - II. Upon activation of G-protein, the G-protein-linked receptor dissociates to activate adenylyl cyclase.
 - III. GTPase terminates the activity of G-protein by hydrolysis of GTP.
 - IV. The ratio of G-protein-linked receptor to G-protein is 1:1.
- A III only
 - B II and III only
 - C III and IV only
 - D I, III and IV only

33. Three proteins isolated from a human cell were investigated for their involvement in cell signalling pathways.

	Protein A	Protein B	Protein C
ATPase activity	+	-	-
GTPase activity	-	-	+
DNA binding	-	+	-
Transmembrane domain	+	-	-

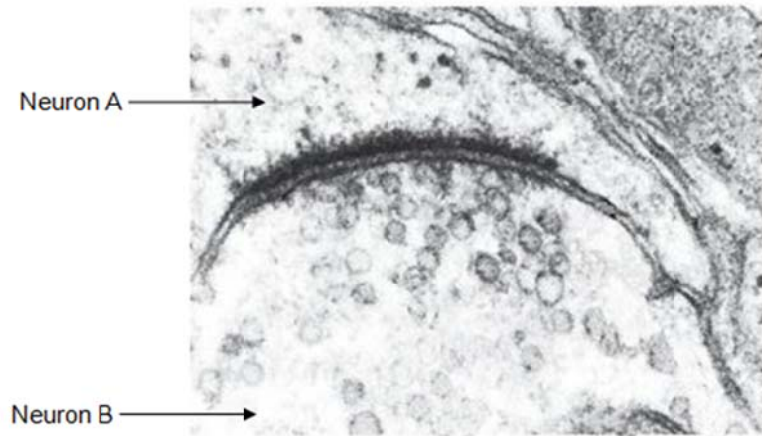
Key: (+) = present, (-) = absent

Which of the following correctly shows the identity of these three proteins?

	Protein A	Protein B	Protein C
A	Na ⁺ -K ⁺ pump	Ras protein	Testosterone receptor
B	Testosterone receptor	Ras protein	Na ⁺ -K ⁺ pump
C	Na ⁺ -K ⁺ pump	Testosterone receptor	Ras protein
D	Ras protein	Testosterone receptor	Na ⁺ -K ⁺ pump

34. How is an increase in the strength of a stimulus communicated by a neurone?
- A** The action potential travels faster along the neurone.
 - B** The duration of the depolarisation phase of the action potential increases.
 - C** The frequency of action potentials generated along the neuron increases.
 - D** The spike of the action potential reaches a higher voltage.

35.



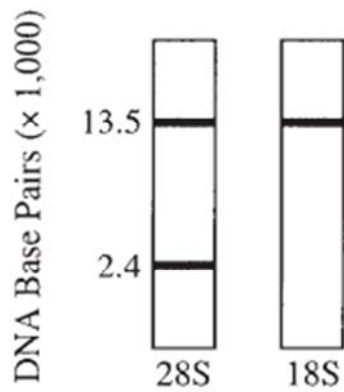
The micrograph above shows a synapse between two neurons. Which of the following statements about synaptic transmission is correct?

- A Neuron B has received the synaptic vesicles from Neuron A.
- B The membrane of Neuron A has many neurotransmitter channels.
- C Calcium influx is triggered by depolarization that is produced by the arrival of an action potential in Neuron B.
- D Neurotransmitters triggers voltage-gated sodium ion channels in Neuron A.

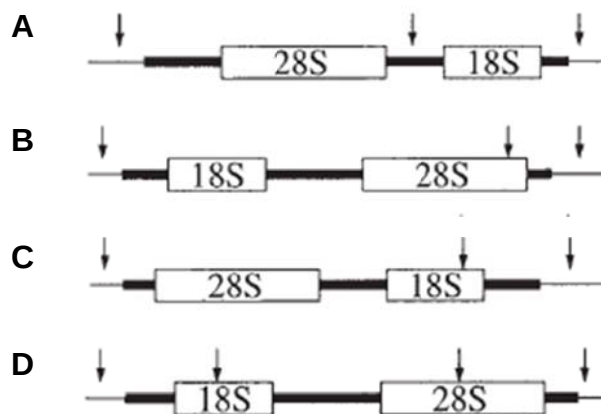
36. Which of the following aims would the creation of a genomic DNA library help to achieve?

- A to determine the sequence of telomeric repeats in a person
- B to identify genes that are responsible for liver cell function
- C to obtain the complete sequence of ras gene
- D to study the changing patterns of gene expression during pig development

37. The autoradiograms obtained below (after electrophoresis and Southern Blotting) show human DNA digested with a specific restriction enzyme and probed with labeled rRNA. In the autoradiogram on the left, the probe was 28S rRNA; at the right, the probe was 18S rRNA.



If the arrow in the following map shows the location of the restriction sites of this restriction enzyme, which map best explains the result shown above?



38. Genetically engineering techniques may be used to mass produce human proinsulin using E.coli as the expression system.

Some events that take place during the process are listed below.

- I Insert the plasmid vector into a plasmid-free E.coli.
- II Use a restriction endonuclease to cut the plasmid vector at specific points.
- III Purify proinsulin from the E.coli.
- IV Grow the E.coli in an industrial fermenter.
- V Insert the proinsulin gene into the plasmid vector.

Which sequence of events is correct?

first → last

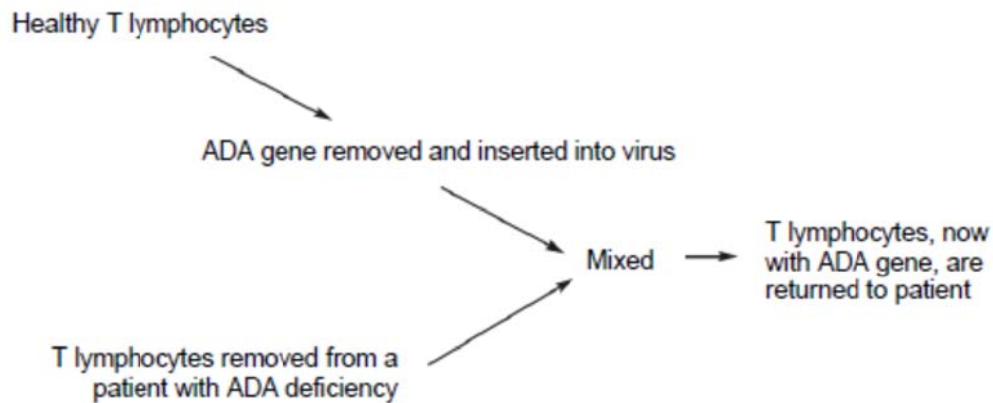
- A** II, I, V, III, IV
 - B** II, I, V, IV, III
 - C** II, V, I, III, IV
 - D** II, V, I, IV, III
39. Among the steps that plant biotechnologists employed when they made the corn plant resistant to the European corn borer was the use of stage A followed by stage B .

Which of the following pairs of statements correctly describe stages **A** and **B**?

	Stage A	Stage B
A	Bt toxin gene isolated from <i>Bacillus thuringiensis</i> with the aid of restriction exonucleases	Bt toxin gene is introduced into corn plant cell via microinjection
B	Bt toxin gene isolated from <i>Bacillus thuringiensis</i> with the aid of restriction endonucleases	Bt toxin gene is introduced into corn plant cell via electrophoresis
C	Bt toxin gene isolated from <i>Bacillus thuringiensis</i> with the aid of restriction exonucleases	Bt toxin gene is introduced into corn plant cell via gene gun
D	Bt toxin gene isolated from <i>Bacillus thuringiensis</i> with the aid of restriction endonucleases	Bt toxin gene is introduced into corn plant cell via electroporation

40. Gene therapy is used to treat a genetic disorder which results from a deficiency of the enzyme, adenosine deaminase(ADA). Without this enzyme, T lymphocytes have impaired function and the immune system of the affected individual is compromised.

The diagram shows the processes involved in the treatment of ADA deficiency by gene therapy.



If the therapy failed in the first round itself, which one is not a possible explanation for the failure?

- A modified T lymphocytes were removed by the patient's immune system
- B ADA was expressed in very low amounts
- C expressed ADA failed to fold in the correct conformation
- D insertion of the ADA gene into the enhancer region of the viral genome